

UNIVERSITY OF ALABAMA IN HUNTSVILLE
INDUSTRIAL & SYSTEMS AND ENGINEERING MANAGEMENT DEPARTMENT
ISE 430/530: MANUF SYS & FACILITIES DESIGN

Fall 2026

Instructor:	Dr. Cheng Chen	Office Hour:	T 3:30-4:30 pm, F 800:-9:00 am
Email:	Cheng.chen@uah.edu	Office:	OKT N133
Class:	TR 01:00 - 02:20 PM	Classroom:	OKT N152

Objectives: Throughout the semester, students will develop an understanding of modern manufacturing systems design, with an emphasis on facility location and plant layout. This includes classical systems, just-in-time systems, and principles of integrated manufacturing systems design, as well as an analysis of process flow, productivity, and available space to determine facility layout:

- Compare and explain the relationships among manufacturing system design, production variety and volume, and material handling requirements.
 - Identify and describe key technologies that support modern manufacturing systems, including numerical control machines, robotics, CAD, CAM, and CIM.
 - Explain the functional role of material handling technologies such as conveyors, AGVs, AS/RS, robotics, and CIM within an integrated manufacturing environment.
 - Apply fundamental control strategies to support factory automation.
 - Analyze AGV systems to determine required fleet size, evaluate single direction conveyor systems to establish appropriate spacing, and determine workstation requirements for single station cells.
 - Develop and evaluate systematic layout plans by progressing from a from-to chart to nodal and block layout representations, and assess how layout decisions influence performance in a lean production system.
 - Perform line balancing for both single model and mixed model production lines.
- Implement group technology principles to design a machine cell that reflects lean manufacturing practices, and distinguish between machine cells and flexible manufacturing systems.

Textbooks: You can find the following PDF online as supplementary materials to help your understanding of the course materials (Any year or edition would be fine):

- Tompkins, J. A., White, J. A., Bozer, Y. A., & Tanchoco, J. M. A. Facilities planning. John Wiley & Sons.
- Groover, M. P. Automation, production systems, and computer-integrated manufacturing. Pearson Education India.
- Sule, D. R. Manufacturing facilities: location, planning, and design. CRC press.
- Pahl, G., Beitz, W., Feldhusen, J., & Grote, K. H. Engineering design: a systematic approach. London: Springer. (Optional)

Office Hours:

- Asking for help is not shameful or embarrassing, although it is common to feel anxious when approaching a teacher. My office is a safe space for every person. The act of entering and conversing about nothing in particular often leads to new insights. Though I typically have my door open outside of office hours, I may not be able to meet for a long period of time. However, you are always welcome to drop by.
- Please feel free to ask your questions after class, by appointment, or in the GroupMe chat provided for this purpose. If the times provided are not sufficient, you may set up a meeting time with me through an appointment. Email is the most effective method of contacting me. I will respond to your inquiry within one business day.

Prerequisites: ISE324 - WORK DESIGN and MAE378 - MATERIALS & MFG PROCESS

Grading scale: (430 students only)

- Attendance: 10%
- Homework: 30%
- One in-class test: 30%
- Final project: 30%
- Total: 100%

Grading scale: (530 students only)

- Homework: 30%
- One in-class test: 30%
- Final project/Research paper: 40%
- Total: 100%

Scores	Final Course Grades
90 - 100	A
80 - 89	B
70 - 79	C
60 - 69	D
≤ 59	F

Course Schedule:

The topics listed here are intended to provide general guidance. As the class progresses, the topics covered and the course schedule may change. I will regularly post lecture materials and course updates on the course [GitHub repository](#):

Table 1:

Date	Special	Reading	Class Topic
20-Aug		Ch. 1 (1.1 - 1.7)	Syllabus Overview
25-Aug	No lecture	Ch. 1 & 2	Finish reading Assignment
27-Aug		Ch. 1 & 2	Product, Process, and Schedule Design
01-Sep		Ch. 2 & 3	Product, Process, and Schedule Design
03-Sep		Ch. 2 & 3	Flow Systems, Activity Relationships and Space Requirements
08-Sep		Ch. 3	Flow Systems, Activity Relationships and Space Requirements
10-Sep		Ch. 3	Flow Systems, Activity Relationships and Space Requirements
15-Sep		Ch. 3	Flow Systems, Activity Relationships and Space Requirements
17-Sep		Ch. 3	Flow Systems, Activity Relationships and Space Requirements
22-Sep		Ch. 4 & 6	Personnel Requirements
24-Sep		Ch. 4 & 6	Layout Planning Models & Design Algorithms
29-Sep		Ch. 4 & 6	Layout Planning Models & Design Algorithms
01-Oct		Ch. 4 & 6	Layout Planning Models & Design Algorithms
06-Oct		Ch. 6	Layout Planning Models & Design Algorithms
08-Oct	Fall Break	Ch. 6	Layout Planning Models & Design Algorithms
13-Oct		Ch. 6	Layout Planning Models & Design Algorithms
15-Oct		Ch. 6	Layout Planning Models & Design Algorithms
20-Oct	Mid-term exam		
22-Oct		Ch. 5	Guest lecture
27-Oct	Review proposal due	Ch. 5	Material Handling
29-Oct		Ch. 7	Quantitative Facilities Planning Models
03-Nov		Ch. 7	Facility Design for Warehouse Operations
05-Nov		Ch. 7 & 9 & 10	Facility Design for Warehouse Operations
10-Nov		Ch. 7 & 9 & 10	Facilities Location
12-Nov		Ch. 7 & 9 & 10	Facilities Location
17-Nov		Ch. 8 & 9	Manufacturing Systems
19-Nov	Project due	Ch. 8 & 9	Manufacturing Systems Informatics
24-Nov	Thanksgiving		No lecture
26-Nov	Thanksgiving		No lecture
01-Dec			Project Presentations

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Table 1: (Continued)

Date	Special	Reading	Class Topic
03-Dec			Project Presentations

Homework: There are up to six homework assignments in total, although the final number will depend on the pace and progress of the course. Each assignment will contain a varying number of questions, and four questions will be randomly selected for grading.

Graduate students enrolled in 530 will be required to complete additional problems for each homework assignment and exam. If a question involves a simulation, you must submit the corresponding code along with your solutions.

All homework submissions must be completed through Canvas. Late submissions will receive a 20% grade penalty.

Final Project: Students may choose your own project as your final evaluation for the course. The project must be related to the general subject of the course. A proposal is required and due Oct 28th, if it involve collection data to generate a from-to-chart, you must included in the report (10 points). A final report will be due on Nov 20th and required to give a presentation.

Tests:

- One in-class tests will be given. Depending on the progress of the lectures, the actual exam date may change.
- No make-up tests will be given. This applies to both excused and unexcused absences. If you miss a test and have a valid excused absence, the points associated with the missed test will be shifted to the final exam. If you do not have a valid excused absence, a grade of zero (0) will be recorded for the test.
- Calculators. Standard scientific calculators are allowed for use during all quizzes. Programmable calculators and wireless devices(e.g., cell phone, iPod/Pad, etc.) are not permitted.
- Formula sheet. You are permitted to bring an A4 formula sheet/a letter size paper to the exam, with notes on both the front and back. Throughout the semester, prepare this sheet by noting down the formulas you deem most important.

Research Paper: 530 students may choose to write a review paper in a selected discipline. The paper topic must be proposed and approved by October 10th to avoid a 10-point deduction, and the completed paper is due by November 12th. The paper should follow the [PRISMA flow diagram](#) for structure. The quality of the paper will directly impact your score. The International Journal of Production Research journal template can be found in the Canvas folder.

Plant Layout: There will be a facilities layout project that will require you to be thinking of a process where you could improve the layout. It can be something from work, a case study that you read about, or a place that is familiar to you.

Academic Coaching: Academic Coaching at the UAH exists to address the holistic needs of undergraduate students that impact their academic success. (Visit <https://www.uah.edu/ssc/tutoring/academic-coaching>)

Presentation Help: UAH students looking for assistance with practicing presentations or public speaking can visit the Presentation Collaboratory. (Visit <https://www.uah.edu/ssc/resources-for-students>)

Student Success Workshops: Student Success Center (SSC) are offered to provide a comfortable environment where students can feel free to ask questions and engage with faculty, academic coaches, mentors, and peers. (Visit <https://www.uah.edu/ssc>)

Attendance: Attendance is required (in-person). Students who are absent without an excuse for more than 3 consecutive classes may have their overall score reduced by an additional 1% per missed class, or may be dropped from the course at the discretion of the instructor.

Communication Skills: Written communication is an important aspect of being an engineer. It is important that engineers are able to clearly explain difficult topics to individuals that may not be as knowledgeable on the subject. Each submittal will be graded in part on your communication skills. These include: spelling, grammar, punctuation, and clarity of writing, enunciation, voice projection, clarity and logical order of presentations.

Professionalism: Engineering faculty at UAH expect students to act in a professional manner at all times, develop the work ethics required for a successful engineering career and follow the Engineering Code of Ethics. Engineering students at UAH are responsible for maintaining the highest standards of professionalism and professional practice.

Inclement Weather: Class will be held at its regularly scheduled time unless the University is shut down or we (instructor) make a personal decision to cancel class. In the event that we cancel class, we will post an e-mail/CANVAS message.

Disability Statement: The University of Alabama in Huntsville will make reasonable accommodations for students with documented disabilities. If you need support or assistance due to a disability, you may be eligible for academic accommodations. [Apply here](#) or contact [Disability Support Services](#) (256.824.1997 or Wilson Hall 128) as soon as possible to coordinate accommodations.

ABET outcome mapped to this course:

OUTCOME (1) ASSESSMENT: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

This is achieved through a plant layout case study. This is one of three plant layout assignments and counted 30-40% of the grade. One of the versions is given below the table. Students were also given the option of coming up with a project and I had one student to take this option.

You will need to make and identify many assumptions as part of the design process. You may also need to perform several iterations of the various design process steps. Please document all assumptions and iterations. I am more interested in the process followed than in the final answer obtained. A project that would be considered “very good” will result in a grade in the lower 90%; to achieve a near 100% you must go beyond what is asked and enter into the realm known as “excellent.” I will be looking for innovation and an attention to details. It will be in your best interest to keep your design to yourself (For example if several people have almost the same “good” or “very good” solution I will take this as an average grade and all will receive 70%.- 80%)

Mental Health Statement: As a student you may experience a range of issues that can cause barriers to learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance or reduce a student’s ability to participate in daily activities.

The University of Alabama in Huntsville offers services to assist you with addressing these and other concerns you may be experiencing. If you or someone you know are suffering from any of the aforementioned conditions, you can learn more about the broad range of confidential mental health services available on campus via the [Department of Student Affairs](#) located under the [Health and Wellness](#) or the UAH Counseling Center by calling 256.824.6203.

24 hour emergency help is also available through the 24/7 National Suicide Prevention Hotline at

1.800.273.TALK or at suicidepreventionlifeline.org or a student who lives on-campus can reach out to the UAH PD dispatch to contact an on-call counselor by calling 256.824.6596. If you find yourself in a mental health emergency, call 6911 on-campus or 911 off-campus.

Subject to Change: Every effort is made to follow the guidelines in the syllabus; however, if needed, the syllabus will be amended. You will be notified if changes are made.